Unit - III

PLD Architecture and Applin

Date
Page

* Topics

- Design Flow. CPLD Architectures, features, Specifiecations, Applin, FPGA Architecture, Features, Specificcations, Applin, Clock management techniques.

- The Simulation & Syntheis Tools, FPGA Synthesis & Impleme^n, Comparison CPLD & FPGA.

PLD - Programmable logic devices

- PLD is a programmable logic device that can be programmed according to the user requirements.

- If discrete IC aee used instead of PLD, a laege amount of space is required. beaause many discrete Ic's are needed to complete the design.

- PLD has many discrete IC's Combined which aee integrated into Single IC That can be programmed according to the user's requirement.

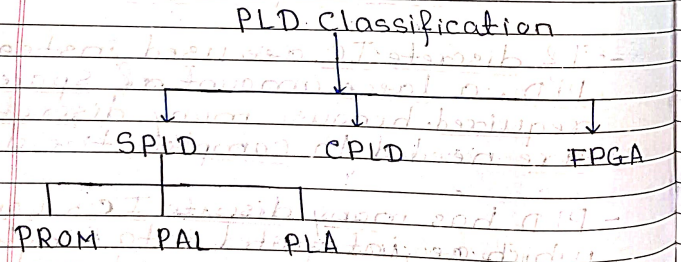
- There are different Softwaee tools availble. By making use of those goods tools, we can reprogram this again

and again.
- We can verify the design using software techniques.

* Adv

- less design time req.
- Low power requirⁿ
- Reduced Space.
- High design Security.
- Low prodⁿ Cost.
- The design is flexible.
- high Switching Speed.

Classificⁿ of PLD's



① SPLD → Simple Programmable logic device
② CPLD → Complex Programmable logic device
③ FPGA → Field Programmable Gate Array

① SPLD
- It has a fixed number of funⁿ.
- limited of logic gates.
- It is a low cost & pin-to-pin performance device.

i) PROM
- Programmable Read only memory
- array of logical AND & OR gates, implemented on a Single chip.

ii) PAL
- Programmable Array Logic
- AND array is programmable, & OR Array is fixed

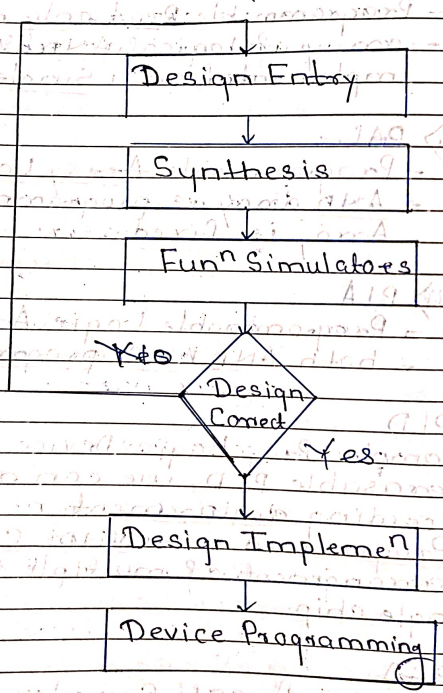
iii) PLA
- Programmable Logic Array.
- both AND & OR programmable.

② CPLD
- Complex Prog. Logic Device
- erasable PLD, we can reprogram according requirement.
- Programmed with VHDL Coding techn.
- arrangment of multiple SPLD's on a Single chip.

③ FPGA
- Suitable for all Complex applications.
- wide density range, thou. to millions logic gates.

PLD Design Flow

- The design Flow means we have to start from a certain higher level by entering the Specifi^{n} of the appl^{n} we want to design.



Design flow

① Design Entry:

- We have to enter the Specification of Things that we want to design.
- These are things related to the particular appl^{n} to designed.
- Create design using
  - HDL - [Hardware descriptive Lang].
  - Schem Entry

② Syntheis

- used for Synthesis purposes,
- which means it converts the System from high level to low-level Circuitory.
- It Identifies the Syntax error if you are using HDL, after errors are Identified, correct them.

③ Functional Simulators.

- The desired Specfications are checked by using the Fun^{n} Simulator Block
- If There is an error, Then Those error are again checked & Corrected.
- If there is errors, design is not Corrected, Then again we have to go back to design Entry

4. Design Implem^{n}

- At this Stage Particular app^{n} is implemented using CPLD or FPGA
- Three approach.
  ① Top-down approach.
  - we have to move top design to the bottom.
  ② Bottom-Up approach.
  - Start low level Compo. design up to the high level design which is final design.
  ③ Mixed Approach.
  - use of both approaches.

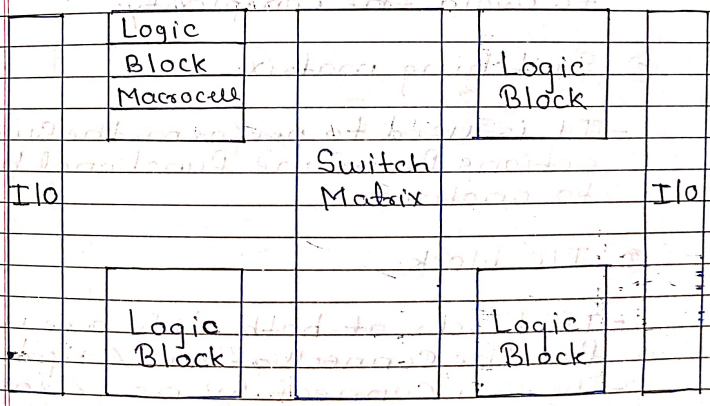
5. Device Programming.

- Generate the binary File PLD.
- Particular appl^{n} is Implemented using CPLD or FPGA
- The Programmed device in real system.

CPLD

- Complex Programmable logic device.
- If the app^{n} is complicated, then depending upon the requirement it is necessary to use a number of SPLD's in a single IC package.
- i.e. Called CPLD.

Architecure of CPLD.



- The CPLD Arch. consists of I/O block, Logical Block, macrocell & Switch matrix block.
- There are 4 I-O.

① Logic Block.

- Logical Block is also called Fun^n Block.
- Fun^n block Consists of many PLD's
- Each PLD is called a macro Cell.
- 'n' no of macro cell in a logic block
- All these logical blocks are interconnected with each other. interCon^n is not shown in this diagram just to avoid the Complexity.

② Switching matrix.

- It is used to perform the Switching actions from one functional block to another.

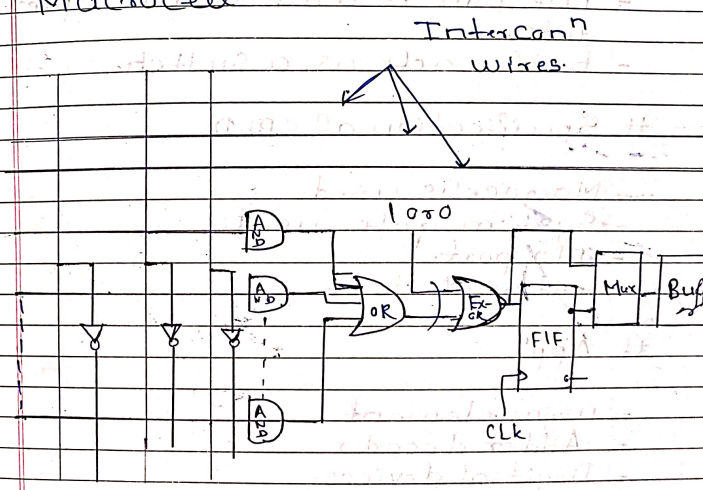
③ I1O block.

- I1o blocks at both ends are used for the Connec^n of I1o & o/p devices.
- we can Connect 'n' no of inputs & outputs.

Features of CPLD.

① device is reprogrammable.
② high Performance.
③ Non-Volatile memory.
④ built in processor.
⑤ you want to reprogram the device you don't have to disconnect it.

Macrocell



- A Functional block has many micro cells all Connected internally.
- Each Microcell has AND Gate, OR gate, XOR Gate,
  AND Gate ⟶ Create product term.
  OR Gate ⟶ Combines the product Term.
  XOR Gate ⟶ it has o/p of Exact

Gate and other input has 1 or 0
- The o/p of the Ex-OR gate is conn ted to the Flip-flop.
- The o/p of the Flip-Flop is connec to Mux and a buffer.
- The Mux is used to multiplex the data.
- buffer acts as a switch.

Specification of CPLD

- Macrocells used.
- System gates used.
- I/o port.
- external clock.

App^n

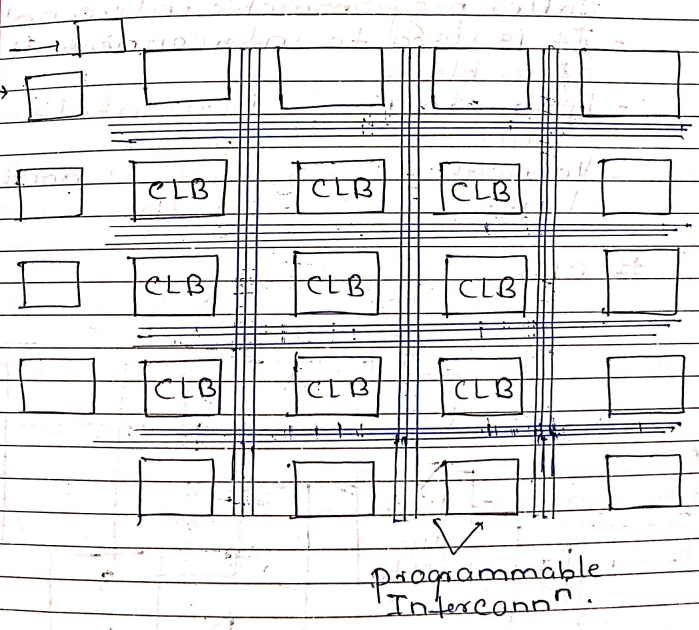
- mem element
- Adder decoder
- Digital devices.

Product Term Allocator

- The macro cell has ALU. The o/p of AND gates are the product term
- To properly control the product term the Product Term Allocator is used

FPGA

- Field Programmable Gate Array.
- It applicable for designing Complicated Circuits.
- It can be reprogrammed.



⊕ I/o Blocks.

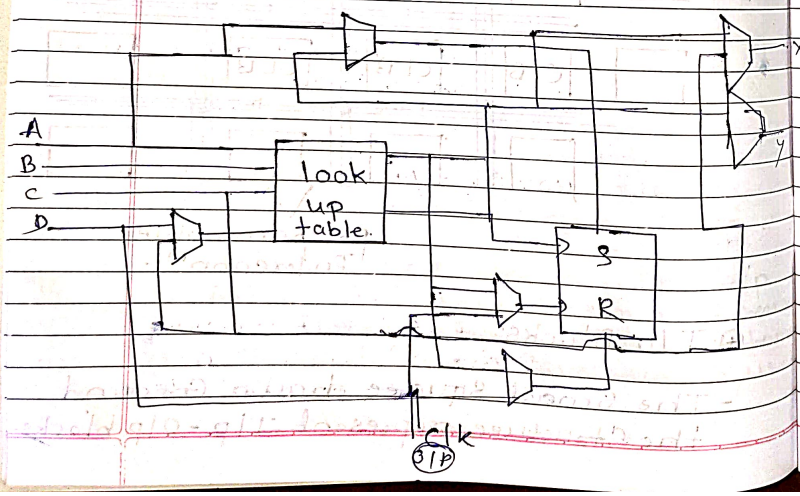
- The Small Square drawn Around the Structure represent I/p - o/p blocks.

- It Connect the external pins to the internal logic.

② Programable Interconnection

- The Vertical and horizontal lines are called programmable interconnn.
- It is used to interconnect CLB's to other block.
- horizontal lines → horizontal routing paths.
- Vertical lines → Vertical routing paths.

CLB



- It Consists lookup tables. The lookup tables are TT
- Implement Seq. Ckt by writing a truth table.
- SR flf is used as a memory element to store the data.
- four inputs (A, B, C, D) are used

1. Lookup tables

- LUT are used for implementing Combinational at logic.
- each LUT stores TT.

2. Carry logic.

- Stores the carry generated by arithmatic Such as addn, sub.

3. Multiplexer
- The multiplexer are used to Multiplexing of the o/p of LUT & FIF.

4. Logic cell.

- The logic cell is the fundamental building Block of an FPGA.
- The Primary Consist of look-UP table
- & flf.

- LUT implements combin^{n} logic fu
- FIF stores the o/p for seq. logic

② Peogrammable Switch matrix.

- The Programmable Switch matrix e
for flexible Conn^{n} with in FPGA.
- acts as switch.

③ I lo Block

- The I lo Block manages the Comm
bet^{n} the FPGA and external devic

features.

① Easy Configu^{n} FPGAs.
② High Speed.
③ reprogrammable.
④ design cycle is short.
⑤ Relability
⑥ Can handle the complicated fun^{n}.

Specification

- Depends on, No of System gates & CLB in the FPGA.
- No of LUT Present
- No of user I lo Pins.
- No of shift register availble.

Need

- To implement complicate ckt
- for reProgramable.

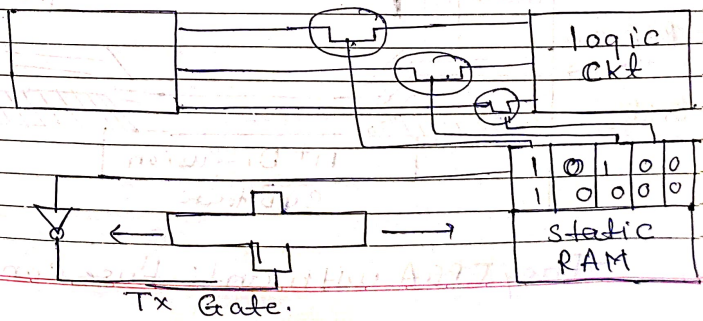
App^{n}

① Digitial Signal Processing
② Image & Video Processing.
③ Automative app^{n}.
④ Aerospace & defence.
⑤ medical equipment.
⑥ wireless Comm^{n}

Technologies Supported by FPGA

① SRAM Tech
② Anti-fuse Tech.

① SRAM
- Static random Access Mem.

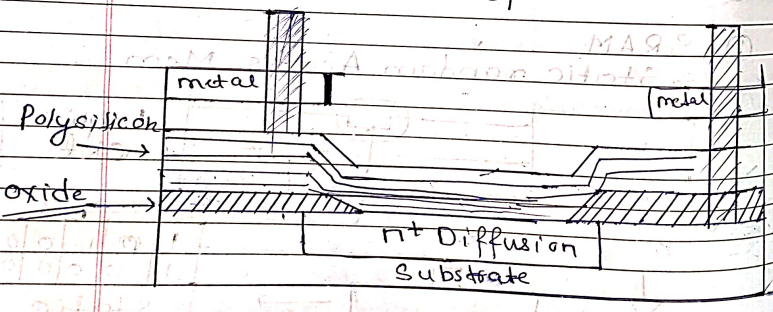


- There are two logic ckt are used.
- In betⁿ two logic ckt N-MOS is connected.
- This mos acts a switch
- The upper N-MOS is connected to logic 1 bit of SRAM, so path is Created.
- The 2nd N-MOS connect to logic 0 of SRAM, so path is ~~Not~~ Not Created.

Adv

① Can be reprogram
② Can be modified as per user req.

② Antifuse Technology.



- They FPGA with anti-fuse connⁿ Programed durning manfactures.

working

- The anti-Fuses connect the logic blocks and interConnect based on the user design.
- Initially, the impedance of the devi is very high. So apply Vtg is lo
- By applying the Vtg, a high imped ance is reduces and the connⁿ can be made.
- Connection made due to the new channel is formed and current start flowing
- The channel is form at threeshold Vtg.
- Adv → require less are.
  - low ON Resistence.
  - low possible Capacitence.
- Disadv → It is NOT Programmed.

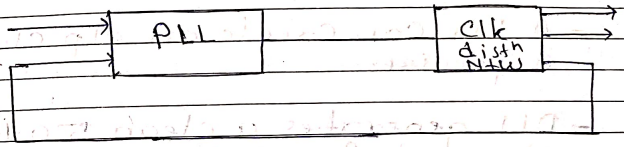
Compare FPGA & CPLD.

feature	CPLD	FPGA
1. Stand for	Comp. Pro. logic	Field Progra
2. Arch.	Macrocell	Look up table
3. Logic Capacity	Low to medium	High to Ultra High density
4. Config Memory	Non-Volatile	Mostly volatile
5. Start-up Time.	Instant on	Slow.
6. Timing Predictability	Highly Predictable	Less Predictable.
7. Flexibility	lower.	Higher.
8. Power consumption	Lower static Power	Higher static Power
9. Applin	Simple state Machines, Address decoding.	DSP, High-Speed Networking, Video/Image Processing.

clock Management Tech FPGA.

① PLL

- Phased locked loop



- The Block diagram shows the use of PLL for clk management.
- PLL generates an Internal clock
- It generates a high freq. clock from external clock.
- It provide freq multiplicator
- Ref clock is applied as one of the input the PLL.
- The o/p of PLL applied to the clock distribution ntw.

② DLL

- Delay locked loop.
- we have a DLL that is delay locked loop.
- It align o/p clock Pulse with the

the reference clock pulse.

- It removes the delay introduced by the internal FPGA clock distribn ntw
- A PLL can create an o/p clock with a precise.
- PLL generates a clean 50% Duty cycle clock for internal logic.

Simuln & Syntheis Tools:

Tools

1. ModelSim → most widely used HDL Simuln tool.
2. QuestaSim → Adv Simuln Tool for large design with ...
3. Verilator → Open-Source Verilog Simulator. Very fast cycle accurate Simuln.
5. Synopsys Vcs → high Speed Simuln Tool maily used for ASIC Verifn.

Syntheis Tools.

1. FPGA Express
   - It is pc based tool used for logic Synthesis & optimn
   - creates netlist for VHDL & Verilog HDL Code.
2. Synopsys Design Compiler
   - Converts Verilog/VHDL Code into a gate-level netlist for ASIC design
3. Intel Quartus Synthesis:
   - Generates FPGA, hardware structure and optimizes logic & timing for Intel devices